COMP5565 Decentralized Apps Fundamentals and Development

Group Project (15%) — 2026/27 Semester 1

Title: "Decentralized Product Provenance and Warranty dApp"

Background

This project focuses on developing a decentralized application (dApp) that leverages blockchain to **record the manufacturing provenance and post-purchase warranty lifecycle** of **high-value consumer electronics**, e.g., luxury smartwatches or smartphones.

Each product will have a record representing its identity, ownership history, and warranty details. The project aims to demonstrate how blockchain can provide **transparency**, **security**, and **trustless warranty management** between manufacturers, service centers, and consumers.

The dApp should be deployed on **Ethereum (public blockchain)** or **Hyperledger Fabric (private consortium network)**, based on the desired privacy model.

System Requirements

Your dApp should allow different stakeholders to interact with the system based on their roles:

1. Stakeholders & Roles

- **Manufacturer:** registers new products and issues initial ownership and warranty details.
- **Retailer:** transfers ownership to the buyer upon sale.
- **Customer:** verifies ownership, requests warranty service, transfers ownership (e.g., resale).
- **Service Center:** validates warranty claims and logs servicing actions.

2. Core Functionalities

A. Product Lifecycle Traceability

Each product (e.g., smartwatch) gets a **unique digital record** stored on-chain with the following:

- Product ID, serial number, model details
- Manufacturer details and timestamp
- Ownership transfers (manufacturer → retailer → customer → resale)
- Warranty details (start date, expiration, claim count)

B. Smart Contract–Based Warranty Management

- Warranty terms (duration, number of claims) encoded in smart contracts
- Customers submit warranty requests via front-end
- Service centers approve/reject warranty claims (logged on-chain)
- Warranty automatically expires after duration or claim limit is reached

C. Ownership Verification Interface

Consumers can verify product authenticity and ownership history
D. Privacy Control
To this end:
1. Outline the architecture of the dApp, including smart contracts, user interfaces, and database design. Note that you will likely need to make compromise on the transparency vs. privacy criteria.
2. Implement your solution according to your architecture. Ensure smart contracts are secure and optimized for gas usage (if Ethereum is chosen).
3. Develop the frontend to interact with the blockchain, directly or via a centralized backend.
4. Deploy smart contracts on a local Ethereum blockchain (Hardhat) or set up a network (Hyperledger Fabric).

Project Deliverables

Part 1: System Design (40 marks)
Prepare a design report describing:

1. System Architecture
2. Security Analysis
3. Accessibility & Usability
4. Efficiency & Cost Optimization
5. Installation and Deployment Guide

Report length: ≤ 30 pages including all diagrams and screenshots; with single line spacing and font size 12. PDF only; APA or IEEE citations.

Part 2: System Implementation (40 marks)
Implement your dApp according to the design:

- **Smart Contract**: written in Solidity (for Ethereum) or Chaincode (for Fabric)
- **Frontend**: For the front-end of your applications, it can be desktop software (GUI/Terminal-based), mobile app, website or a combination of them.

Part 3: Presentation (20 marks)

- 10~12-minute in-class presentation on the last lecture
- Key Components:
    - System architecture overview
    - Smart contract demonstration (basic interaction)
    - Live walkthrough of ownership transfer or warranty claim

(Presentation schedule to be released later.)

Remarks

- **Group size:** 5 students
- **Group leader:** to register group by **1st November 2026**
- Focus on *achievable*, working smart contract logic rather than full enterprise deployment
- Creativity in **UI design** and **data visualization** is encouraged

Assessment Criteria

This project is assessed based on the following rubrics:

	Excellent	Good	Satisfactory	Weak	Fail
System Design (40 marks)	Identify all requirements and provide justifications. The design is in a well-structured and detailed manner. (31-40 marks)	Identify most of the requirements and provide justifications. The design is comprehensive. (21-30 marks)	Identify some of the requirements and provide justifications. There are minor flaws in the design. (11-20 marks)	Identify a few requirements and provide justifications. There are critical flaws in the design. (1-10 marks)	Unable to identify the requirements, and there are no proper justifications and design. (0 mark)
System Implementation (40 marks)	Develop an application that satisfies all the system requirements. (31-40 marks)	Develop an application that satisfies most of the system requirements. (21-30 marks)	Develop an application that satisfies some of the system requirements. (11-20 marks)	Develop an application that satisfies a few of the system requirements. (1-10 marks)	Unable to develop an application that satisfies the system requirements. (0 mark)
Presentation (20 marks)	The presentation has an attention-grasping Introduction, well organized, coherent, includes a clear demonstration, excellent time management and visual aids. (16-20 marks)	The presentation has an introduction of the design and clear demonstration, coherent, good time management and visual aids. (11-15 marks)	The presentation has an introduction of the design and demonstration, fair time management and visual aids. (6-10 marks)	The presentation has poor introduction of the design and demonstration, uses time poorly. The group has trouble bringing visual aids smoothly into the presentation, it is illogical. (1-5 marks)	The presentation does not have an introduction of the design and demonstration, no time management and no visual aids, it is illogical. (0 mark)

Submission

The submission deadline of the system, report, and slides is **23:59:00 1st December 2026**. No late submission is allowed.

Only the group leader has to submit the followings:

Part 1 – Use PDF format for the report, and PDF or PPTX for the slides. Name the files using the following convention:

Group<group_no>-report.pdf, e.g., Group99-report.pdf
Group<group_no>-slides.pptx, e.g., Group99-slides.pptx
Group<group_no>-slides.pdf, e.g., Group99-slides.pdf

Part 2 Put all of your source codes into a single folder (which can contain a number of folders/subfolders). The files/folders must be organized properly that can be easily deployed, based on the installation guide in Part 1.

Note: There is no need to submit the blockchain framework related binaries (e.g., node_modules, and dockers for the Hyperledger Fabric network). Make sure you provide the detailed installation guidelines for your applications in the report.

Zip the folder using the following naming convention:

Group<group_no>.zip, e.g., Group99.zip

Submit the ZIP file to Blackboard. If the ZIP file is too large, put it to your PolyU OneDrive storage. Save the link in a text file and submit the text file.